Requirements Writing for AI Procurement and Implementation

PGM 210

AI procurement requires stronger requirements than ordinary software purchasing because agencies must define not only features, but also data rights, validation, interoperability, security, privacy, monitoring, documentation, accessibility, and exit strategy. This module teaches learners how to write requirements that can be evaluated, tested, enforced, and monitored throughout implementation.

Level	applied/foundational
Primary track	Program Management Role-Based Track
Appears in tracks	operations-data-quality

Learning Objectives

- Explain the purpose of requirements writing for ai procurement and implementation in public health AI work.
- Identify the technical, operational, privacy, security, equity, and governance issues associated with the topic.
- Apply the topic to a real or realistic public health workflow, system, or implementation scenario.
- Recognize common failure modes that can affect safety, accuracy, trust, workflow fit, or sustainability.
- Identify practical guardrails that should be documented before implementation.
- Produce a AI requirements matrix that can support readiness assessment, governance review, procurement, implementation, or monitoring.

Module Overview

AI procurement requires stronger requirements than ordinary software purchasing because agencies must define not only features, but also data rights, validation, interoperability, security, privacy, monitoring, documentation, accessibility, and exit strategy. This module teaches learners how to write requirements that can be evaluated, tested, enforced, and monitored throughout implementation.

Jurisdiction and Agency Policy Note

This module provides national-level training guidance for public health agencies and partners. It is not legal advice and does not replace review by agency legal counsel, privacy officers, procurement officials, civil rights staff, records officers, or other authorized decision-makers. Requirements may vary by state, locality, territory, Tribe, agency, funding source, data-sharing agreement, emergency authority, and organizational policy. Learners should use this module to identify the issues that require review and should confirm applicable requirements before approving, procuring, deploying, or publicly communicating about AI-supported work.

Why This Topic Matters for Public Health AI

Requirements Writing for AI Procurement and Implementation matters because AI systems are embedded in public health programs, data systems, and decision processes rather than operating in isolation. The topic affects whether AI outputs are accurate, explainable, safe, equitable, secure, and sustainable. Agencies that ignore this area may create tools that work in a demonstration but fail in actual practice.

The topic also affects accountability. Public health agencies must be able to explain how data are used, why a tool was approved, what evidence supports the workflow, who reviews outputs, and what happens when the system fails. These responsibilities become more important as AI tools move from experimentation into operational use.

Core Concepts

A requirement is a clear statement of what a system must do or how it must perform. AI requirements must cover functional capabilities, nonfunctional expectations, data handling, validation, transparency, monitoring, security, accessibility, interoperability, and support.

Functional requirements describe what the tool must do, such as summarize a case report, classify a complaint, or generate a draft message. Nonfunctional requirements describe how the tool must perform, such as uptime, response time, security, auditability, explainability, and accessibility.

Procurement requirements must be testable. If a requirement cannot be evaluated during proposal review, implementation, or acceptance testing, it is unlikely to protect the agency. Requirements should avoid vague claims such as “uses responsible AI” unless they are translated into evidence, controls, and deliverables.

Public Health Example

A state health agency procuring an AI-supported case triage tool should require interoperability with the surveillance system, role-based access, source traceability, validation results, subgroup performance reporting, audit logs, human review controls, incident reporting, documentation, and data deletion at contract end. These requirements make vendor claims testable and enforceable.

Topic Deep Dive

Strong AI requirements begin with intended use. The agency should state the specific workflow, users, data sources, outputs, and decisions the system may support. It should also state prohibited uses. This prevents scope creep and reduces the chance that a tool will be used for decisions it was not designed or approved to support.

Data requirements should specify allowed data sources, minimum necessary fields, terminology standards, data quality expectations, retention, ownership, reuse limits, and deletion requirements. If vendors will use agency data to improve models, that use should be explicitly approved or prohibited.

Acceptance and monitoring requirements should specify what evidence the vendor must provide before launch and throughout operations. This may include validation reports, test cases, model cards, change logs, uptime reports, incident reports, bias assessments, security documentation, and user training materials.

Risks, Failure Modes, and Guardrails

A common failure mode is treating the topic as a technical detail rather than a public health control. When staff do not connect technical decisions to authority, workflow, equity, privacy, security, and accountability, the agency may adopt a tool that appears useful but cannot be sustained or defended.

Another risk is relying on vendor claims without agency-defined evidence. Vendors may provide demonstrations, dashboards, or summary metrics that do not match the agency workflow or data environment. A guardrail is to require documentation, test results, acceptance criteria, and agency-controlled review.

A third risk is failing to plan for change. Public health data, policies, systems, and emergencies change. Any technical approach must include monitoring, review, versioning, escalation, and retirement pathways so the agency can respond when assumptions no longer hold.

Application to AI-Supported Workflows

Generative AI, predictive analytics, RAG, automation, and agentic workflows all depend on the controls described in this module. The controls should be scaled to the risk of the use case. A tool that drafts internal training text may require lighter review than a tool that updates case records, prioritizes outreach, or supports public communication during an emergency.

The module should be applied during readiness assessment, procurement, implementation planning, pilot testing, and post-deployment monitoring. Learners should document how the topic affects data, systems, users, safeguards, and decision-making for the selected workflow.

Reflection Questions

Where does this topic appear in one current or proposed AI workflow in your agency?

What decisions, data sources, users, or partners would be affected if this area is poorly designed?

What evidence would governance reviewers need before approving the workflow?

Which safeguards should be required before pilot testing?

Who owns ongoing monitoring and review?

Practical Exercise

Create a AI requirements matrix for one real or realistic public health AI workflow. The artifact should describe the use case, affected users, data sources, system dependencies, risks, guardrails, review points, and unresolved questions.

The exercise should produce a work product that could be used in a governance meeting, procurement review, implementation planning session, pilot evaluation, or monitoring review. Learners should document assumptions and identify which staff or partners need to be consulted.

Expected Artifact or Evidence

The expected artifact is a completed AI requirements matrix. It should be specific to a public health workflow and should include technical dependencies, governance questions, human review expectations, monitoring indicators, and unresolved risks.

Expected Assignment

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Knowledge Check

- 1. What is the best reason to address requirements writing for ai procurement and implementation before deployment?
- 2. Which practice best supports responsible implementation?
- 3. Which statement best describes a common failure mode?
- 4. What should be included in the practical artifact for this module?
- 5. When should the issue be escalated for governance or leadership review?

References and Resources

- NIST AI Risk Management Framework: https://www.nist.gov/itl/ai-risk-management-framework
- ONC interoperability resources and agency procurement templates
- NIST Generative AI Profile: https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence
- CDC Public Health Data Strategy and Data Modernization resources: https://www.cdc.gov/public-health-data-strategy/php/about/index.html
- OWASP Top 10 for LLM Applications, when security or AI integrations are relevant
- Agency privacy, cybersecurity, data governance, procurement, and records management policies